

Testing & Evaluation

Date	14 July 2026
Team ID	
Project Name	Voice-Based Concept Understanding Analyzer (VBCUA)

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This section summarises how the Voice-Based Concept Understanding Analyzer (VBCUA) was tested and evaluated, drawing on the project's Functional & Performance Testing report and UAT Execution & Report where applicable, and identifying evaluation activities not yet conducted as future enhancements.

Testing & Evaluation	Functional testing: Functional testing covered audio upload validation, transcription behaviour, reference-answer loading, similarity scoring, grade and feedback generation, PDF report generation, and report download. These were structured as nine functional test cases (FT-01 to FT-09) in the project's Functional & Performance Testing report, all of which passed. AI output validation: The two model-driven outputs are validated programmatically rather than left unchecked: the Whisper transcript is type-checked to confirm it is a string before further use (transcriber.py), and the Sentence Transformer similarity score is clamped to the valid 0.0–1.0 range (semantic_analyzer.py) before being converted into a percentage and grade. Beyond this structural validation, no automated check of transcription content correctness (e.g., comparison against an independently verified transcript) is currently implemented. Edge case testing: Edge cases exercised during testing included empty or corrupted audio files, a missing or empty reference answer file, and scores at each grade-boundary threshold (90%, 75%, and 60%). All were handled gracefully with descriptive warnings or errors rather than an application crash, as documented in test cases FT-02, FT-04, and FT-06 of the Functional & Performance Testing report. Performance testing (latency): Response-time and end-to-end latency were measured for the transcription step and the full analysis pipeline (PT-01 and PT-02 in the Functional & Performance Testing report), with indicative completion times of approximately 12–18 seconds for transcription alone and 15–22 seconds end-to-end for a one-minute audio clip on CPU-only hardware. These figures vary with audio length and available hardware. Output quality evaluation: Output quality is currently assessed only through the automated similarity score and resulting grade tier; no formal study comparing VBCUA's grades against independent human-instructor judgments has been conducted. Future enhancement: a calibration study correlating VBCUA's scores and grades with human grading on the same set of spoken responses would strengthen confidence in the tool's assessment quality. User testing & feedback: <i>[Not Yet Conducted]</i> No formal user testing (such as usability sessions with students or instructors) has been conducted on VBCUA to date. The personas, empathy map, and customer journey map
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	developed earlier in the project were analytical design tools used to guide requirements, not direct feedback collected from real users. Future enhancement: a small pilot with actual students and instructors, gathering structured usability and satisfaction feedback, is identified as a valuable next step before wider use. Bug fixing: Defects identified during testing were tracked and resolved as documented in the project's UAT Execution & Report. Of 28 total defects logged, 14 were fixed, with the remainder classified as by-design, duplicate, external, not reproduced, skipped, or won't-fix. Evaluation metrics: The primary evaluation metric used is cosine similarity between the transcript and reference-answer embeddings, converted into a percentage score and mapped to four grade tiers (Excellent, Good, Fair, Needs Improvement). No additional metrics, such as Word Error Rate (WER) for transcription quality or precision/recall for grade classification against a labelled validation set, are currently computed. Future enhancement: introducing WER to quantify transcription accuracy, and precision/recall-style analysis of grade assignment against a human-labelled validation set, would provide a more rigorous, multi-metric evaluation framework.
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